Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

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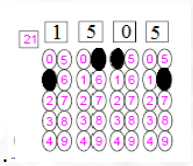
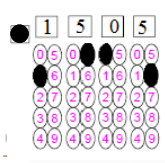
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 25-01-2026**

Time: 09.00Am to 12.00Pm **GTM-19** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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25-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_GTM-19_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. The primitive of $\frac{1}{(x-4)^{3/2}(3-x)^{1/2}}$ is (where 'C' is integration constant) :

1) $\left(\frac{3-x}{x-4}\right)^{1/2}+C$ **2)** $\frac{3}{4}\left(\frac{3-x}{x-4}\right)^{1/2}+C$ **3)** $\left(\frac{x-4}{3-x}\right)^{1/2}+C$ **4)** $2\sqrt{\frac{3-x}{x-4}}+C$

2. If $A=\begin{bmatrix}1 & -1 & 1\\ 0 & 2 & -3\\ 2 & 1 & 0\end{bmatrix}$ and $B=(adj\ A)$ and $C=5A$, then the value of $\frac{|adj\ B|}{|C|}=$

1) 25 **2)** 2 **3)** 1 **4)** 5

3. The solution of $(x+y+1)\frac{dy}{dx}=1$ is

1) $x+y+2=ce^{y}$ **2)** $x-y+2=ce^{y}$ **3)** $x+y-2=ce^{y}$ **4)** $x+y+2=ce^{2y}$

4. The letters of the word COCHIN are permuted and all permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is

1) 96 **2)** 48 **3)** 183 **4)** 267

5. The system of homogeneous equations $\lambda x+(\lambda+1)y+(\lambda-1)z=0$,

$(\lambda+1)x+\lambda y+(\lambda+2)z=0, (\lambda-1)x+(\lambda+2)y+\lambda z=0$ has non trivial solution for

1) Exactly 3 real values of λ **2)** Exactly 2 real values of λ

3) Exactly 1 real value of λ **4)** Infinitely many real values of λ

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6. If $2f(x)+xf\left(\frac{1}{x}\right)-2f\left[\left|\sqrt{2}\sin\pi\left(x+\frac{1}{4}\right)\right|\right]=4\cos^2\left(\frac{\pi x}{2}\right)+x\cos\frac{\pi}{x}\ \forall x\in R-\{0\}$, then which of the following is false ?

1) $f(2)+f\left(\frac{1}{2}\right)=1$ **2)** $f(2)+f(1)=f\left(\frac{1}{2}\right)$

3) $f(2)+f(1)=0$ **4)** $f(1)f\left(\frac{1}{2}\right)f(2)=1$

7. Let $P(\alpha,\beta,\gamma)$ be the foot of the perpendicular from the point $A(3,-1,4)$ on the line $L:\frac{x-2}{2}=\frac{y-3}{-3}=\frac{z+1}{1}$, then $\alpha+\beta+\gamma$ equals :

1) 2 **2)** 4 **3)** 6 **4)** 8

8. The sum of all the elements of set $X=\{x\in\{1,2,.....,100\}:\text{HCF}(x,24)=1\}$ is

1) 1440 **2)** 1536 **3)** 1537 **4)** 1633

9. **Statement –I** : If algebraic sum of length of perpendiculars drawn from points $A(4,2),B(2,4)$ and $C(0,0)$ to variable line is zero, then this line passes through $(2,2)$. and

Statement –II : If algebraic sum of length of perpendicular drawn from three non-collinear points $A(x_1,y_1),B(x_2,y_2),C(x_3,y_3)$ to a variable line is zero, then this line passes through centroid of ΔABC

1) Both Statement –I and Statement –II is true

2) Both Statement –I and Statement –II is false

3) Statement-I is true but Statement-II is false

4) Statement-I is false but Statement-II is true

10. The Area enclosed between the curves $y^2=4x$ and $x^2=4y$ inside the square formed by the lines $x=1, y=1, x=4, y=4$ is

1) $\frac{8}{3}$ **2)** $\frac{16}{3}$ **3)** $\frac{13}{3}$ **4)** $\frac{11}{3}$

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11. Let for a triangle ABC, $\overline{AB}=-2\bar{i}+\bar{j}+3\bar{k}, \overline{CB}=\alpha\bar{i}+\beta\bar{j}+\gamma\bar{k}, \overline{CA}=4\bar{i}+3\bar{j}+\delta\bar{k}$

If $\delta>0$ and the Area of the Δ ABC is $5\sqrt{6}$. Then $\overline{CB}.\overline{CA}$ is equal to

1) 60 **2)** 120 **3)** 108 **4)** 54

12. If Z is the set of integers. Then, the relation $R=\{(a,b):1+ab>0\}$ on Z is

1) Reflexive and transitive but not symmetric

2) Symmetric and transitive but not reflexive

3) Reflexive and symmetric but not transitive

4) An equivalence relation

13. The value of $36\left(4\cos^2 9^0-1\right)\left(4\cos^2 27^0-1\right)\left(4\cos^2 81^0-1\right)\left(4\cos^2 243^0-1\right)$

1) 36 **2)** 27 **3)** 18 **4)** 54

14. Let $\alpha,\beta(\alpha>\beta)$ be the roots of the quadratic equation $x^2-x-7=0$ if

$P_n=\alpha^n-\beta^n, n\in N$ then $\dfrac{P_{20}P_{21}-P_{19}P_{21}-P_{20}{}^2+P_{19}P_{20}}{P_{18}P_{19}}$ equal to

1) 16 **2)** 25 **3)** 14 **4)** 49

15. For a statistical data $x_1,x_2,.....x_{10}$ of 10 values, a student obtained the means as 5.5 and $\sum_{i=1}^{10}x_i^2=371$. He later found that he had noted two values in the data incorrectly as 4 and 5, instead of the correct values 6 and 8 respectively. The variance of the corrected data is

1) 9 **2)** 5 **3)** 7 **4)** 4

16. The line $y=x+1$ meets the ellipse $\dfrac{x^2}{4}+\dfrac{y^2}{2}=1$ at two points P and Q. If r is the radius of the circle with $\overline{PQ}$ as diameter, Then $(3r)^2=$____

1) 20 **2)** 12 **3)** 11 **4)** 8

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17. The function $f(x)=\begin{cases}\left(\frac{4}{3}\right)^{\frac{\tan 4x}{\tan 3x}}; & 0<x<\pi/2\\ b+2; & x=\pi/2\\ \left(1+|\cos x|\right)^{\frac{a}{b}|\tan x|}; & \frac{\pi}{2}<x<\pi\end{cases}$

is continuous at $x=\frac{\pi}{2}$ Then the sum of $(a+b)$ is

1) -1 **2)** 1 **3)** 2 **4)** -2

18. Number of zeros at the end in the expansion of 100! is

1) 22 **2)** 23 **3)** 24 **4)** 25

19. Let T_n be n^{th} term of an A.P. If $T_6+T_{10}+T_{19}+T_{12}+T_{13}+T_{15}=600$, then sum of first 24 terms of A.P. is

1) 1200 **2)** 1800 **3)** 2400 **4)** 3600

20. If f is differentiable function defined over R, $f(5)=7$ *and* $f'(5)=7$, then

$\lim\limits_{x\to 5}\frac{xf(5)-5f(x)}{x-5}$ is given by

1) 35 **2)** -35 **3)** 28 **4)** -28

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. An ellipse has OB as semi minor axis, F and F' are its foci and the angle FBF' is a right angle, and the eccentricity of ellipse is $\frac{\alpha}{\sqrt{\beta}}$ (where α,β are co-prime). Then the value of $\left|\alpha+\beta^2\right|$ is equal to _____

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22. If $\alpha = \int_{\frac{1}{2}}^{2} \frac{\tan^{-1} x}{2x^2 - 3x + 2} dx$, Then the value of $\sqrt{7}\left(\frac{\tan 2\alpha\sqrt{7}}{\pi}\right)$ is _____

Here, The Inverse trigonometric function $\tan^{-1} x$ assumes values in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

23. A bag contains 8 unbiased coins and 2 biased coins with probability of tail is $\frac{1}{3}$. If one coin is drawn from bag and found to be head then the probability that drawn coin is biased coin.is P then 100 P is equal to _____

24. If N is the numerical value of term independent of x in the expansion of $\left(2x^2 - \frac{3}{x}\right)^{15}$. then number of prime factors of N is _____

25. The number of real solutions of the equation $\sqrt{1+\cos 2x} = \sqrt{2}\sin^{-1}(\sin x)$ in $-\pi \le x \le \pi$ is ____

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NEET 2023 BORA VARUN CHAKRAVARTHI
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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. The trajectory of projectile, projected from the ground is given by $y = x - \frac{x^2}{20}$.

Where x and y are measured in meter. The range of projectile is (in meters)

$(g = 10\,ms^{-2})$

1) 5 **2)** 20 **3)** 25 **4)** 2

27. Match List-I with List-II.

List-I	List-II
A. Coefficient of viscosity	I. $\left[ML^2T^{-2}\right]$
B. Surface Tension	II. $\left[ML^2T^{-1}\right]$
C. Angular momentum	III. $\left[ML^{-1}T^{-1}\right]$
D. Rotational kinetic energy	IV. $\left[ML^0T^{-2}\right]$

1) A-II. B-I. C-IV. D-III **2)** A-I, B-II, C-III, D-IV

3) A-III, B-IV, C-II, D-I **4)** A-IV, B-III, C-II, D-I

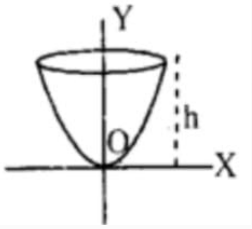
28. A uniform ring and a uniform solid sphere having same mass rotating about an axis passing through their centres have Same radii of gyration. The axis of rotation is perpendicular to plane of ring. The Ratio of radius of ring to that of sphere is $\sqrt{\frac{2}{x}}$. The value of x is _____

1) 3 **2)** 4 **3)** 2 **4)** 5

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29. A paraboloid shaped solid object has equation $y = Kx^2$ about y-axis as shown in figure. If the height of the body is 'h ' then the position of centre of mass from origin (Assume density to be uniform throughout) is

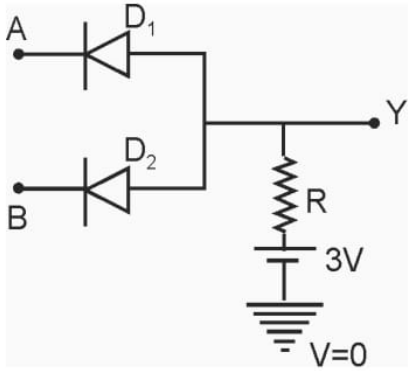


1) $\frac{2h}{3}$ **2)** $\frac{h}{3}$ **3)** $\frac{4h}{3}$ **4)** 2h

30. Earth has mass 27 times and radius 3 times that of a planet. If the escape velocity from the earth is 11.2 km/s, the escape velocity in km /s from the planet will be :

1) 2.8 **2)** 11.2 **3)** 5.6 **4)** 3.7

31. In the circuit, the logical value of A = 1 or B = 1 when potential at A or B is 3V and the logical value of A = 0 or B = 0 when potential at A or B is 0 V. Here Y is the potential at the point Y mentioned in the figure. The truth of the given circuit will be



1)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

2)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

3)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

4)

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

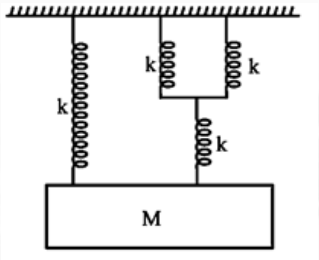
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32. The interference pattern is obtained with two coherent light sources of intensity ratio $4:1$. And the ratio $\frac{I_{max}}{I_{min}}$ is

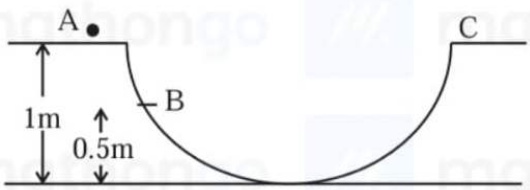
1) 9 **2)** 4 **3)** 2 **4)** 1

33. The time period of simple harmonic motion of mass M in the given figure is $\pi\sqrt{\frac{\alpha M}{5K}}$ where the value of α is



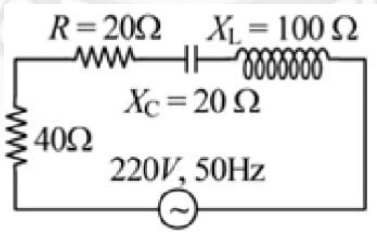
1) 4 **2) 7** **3) 12** **4)** 2

34. A particle is placed at the point A of a frictionless track ABC as shown in figure. It is gently pushed towards right. The speed of the particle when it reaches the point B is: $\left(Take\ \ g=10ms^{-2}\right)$



1) 20 ms^{-1} **2)** $\sqrt{10}ms^{-1}$ **3)** $2\sqrt{10}ms^{-1}$ **4)** $10ms^{-1}$

35. The power factor of the circuit shown in the figure is

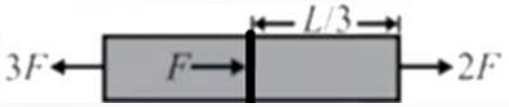


1) 0.6 **2)** 0.1 **3)**1.5 **4)** 2

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36. A uniform cylindrical rod of length L, cross-sectional area A and Young's modulus Y is acted upon by the forces as shown in figure. The elongation of the rod is

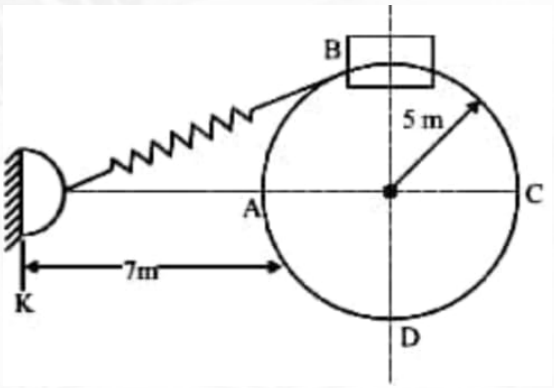


1) $\frac{2FL}{5AY}$ **2)** $\frac{3FL}{8AY}$ **3)** $\frac{3FL}{5AY}$ **4)** $\frac{8FL}{3AY}$

37. Two wires A and B are made up of the same material and have the same mass. Wire A has radius of 2.0 mm and wire B has radius of 4.0 mm. The resistance of wire B is 2Ω. The resistance of wire A is(in Ω)

1) 11 **2)** 22 **3)** 32 **4)** 42

38. A collar B of mass 4 kg is constrained to move along a horizontal smooth and fixed circular track of radius 5m as shown in figure. The spring lying in the plane of the circular track and having spring constant $400\,N/m$ is under formed when the collar is at A. If the collar starts from rest from B, then find the normal reaction exerted by the track on the collar when it passes through A. [Natural length of spring = 7 m]



1) 1400 **2)** 1250 **3)** 2880 **4)** 2250

39. De-Broglie wavelengths of the particle increases by 75% then kinetic energy of particle becomes x times to initial value . Find x ?

1) $\frac{16}{49}$ times **2)** $\frac{9}{16}$ times **3)** $\frac{16}{25}$ times **4)** $\frac{4}{9}$ times

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40. If the distance between object and its two times magnified virtual image produced by a spherical mirror is 15 cm, the magnitude of focal length of the mirror must be:

1) 15 cm **2)** 12 cm **3)** 10 cm **4)** $\frac{10}{3}cm$

41. Assertion(A) : When a monochromatic light incident on a photo electric emitter, all the emitted electrons do not have the same Kinetic Energy.

Reason (R) : In a monochromatic light, different photons have different wavelengths.

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true but R is not the correct explanation of A

3) A is true but R is false

4) Both A & R are incorrect

42. Assertion: Electromagnetic waves are transverse in nature.
Reason: The electric and magnetic fields of an e.m. wave are perpendicular to each other and also vibrating perpendicular to the direction of wave propagation.

1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

2) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.

3) Assertion is true, Reason is false.

4) Assertion is false, Reason is true.

43. Given below are two statements:
Statement I: In a diatomic molecule, the rotational energy at a given temperature obeys Maxwell's distribution.
Statement II: In a diatomic molecule, the rotational energy at a given temperature equals the translational kinetic energy for each molecule.

In the light of the above statements, choose the correct answer from the options given below:

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1) Statement - I is false but Statement - II is true.

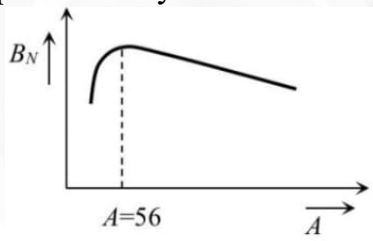
2) Statement - I is true but Statement - II is false.

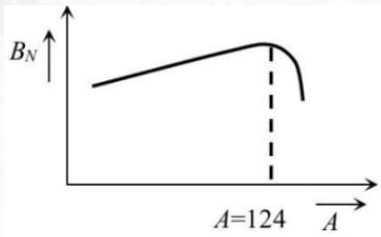
3) Both Statement - I and Statement - II are true.

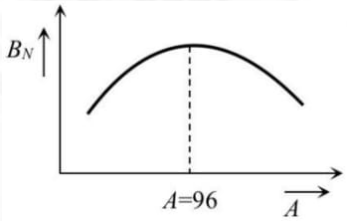
4) Both Statement - I and Statement - II are false

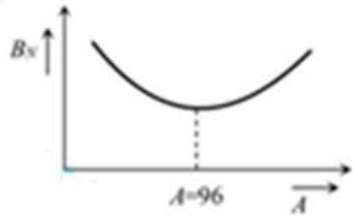
44. Given below are two statements
Statement (I): Viscosity of gases is greater than that of liquids.
Statement (II): Surface tension of a liquid decreases due to the presence of insoluble impurities.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Statement I is correct but statement II is incorrect
2) Statement I is incorrect but Statement II is correct
3) Both Statement I and Statement II are incorrect
4) Both Statement I and Statement II are correct

45. The dependence of binding energy per nucleon, B_N on the mass number, A, is represented by

1)


2)


3)


4)


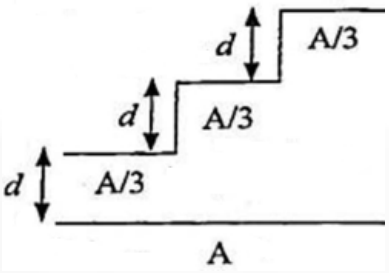
SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A capacitor is made of flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is A/3 and the height is d, the capacitance of the arrangement is $\frac{11A\varepsilon_0}{p\times 3d}$. Find the value of p?

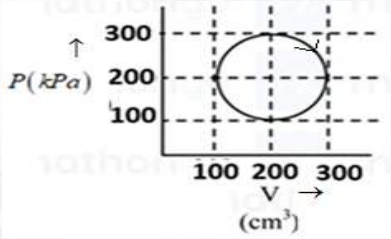
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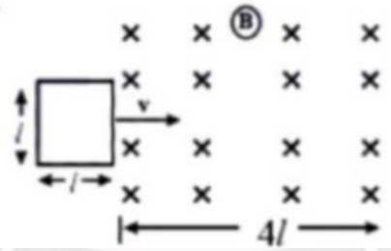


47. The electric field in a region is given by $\vec{E}=\left(2\hat{i}+4\hat{j}+6\hat{k}\right)\times10^3\,N/C$ .The flux of the filed through a rectangular surface parallel to $x-z$ plane is 6.0 Nm^2C^{-1}.The area of the surface is_____cm^2

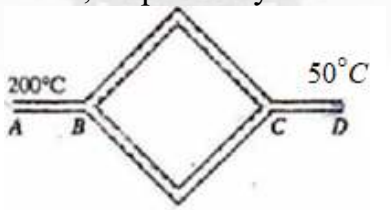
48. An ideal gas has undergone through the cyclic process as shown in the figure. Work done by the gas in the entire cycle is ____$\times10^{-1}$ J . (Take $\pi=3.14$)



49. A square loop of edge l and resistance R enters into a uniform magnetic field B at time t=0. The loop crosses the magnetic field with constant velocity v. If the heat produced in the loop is $\frac{kB^2vl^3}{R}$, the value of k is_____



50. Six identical conducting rods are joined as shown in fig. Points A and D are maintained at temperatures $200^\circ C$ and 50°C, respectively. The temperature of junction B will be



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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Statement-I: 2-pentanone gives positive test with $I_2 / NaOH$

Statement-II: 3-pentanone does not give positive test with $I_2 / NaOH$

1) Statement-I is correct and Statement-II is correct

2) Statement-I is correct and Statement-II incorrect

3) Statement-I is incorrect and Statement-II correct

4) Statement-I is incorrect and Statement-II incorrect

52. The statement (s) that are correct about the species S^{-2}, Cl^{-}, K^{+} and Ca^{+2}

A) All are isoelectronic

B) All have the different nuclear charge

C) S^{-2} has the largest ionic radii

D) Ca^{+2} has the smallest ionic radii

Choose the most appropriate answer from the option given below.

1) (A), (B), (C) and (D) **2)** (B), (C) and (D) only

3) (A), (C) and (D) only **4)** (C) and (D) only

53. Assertion (A) :All isotopes of a given element show the same type of chemical behaviour.

Reason (R) :The chemical properties of an atom are controlled by the number of electrons in the atom.

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) Both A and R are false

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54. Assertion (A) combustion of16 g of methane gives 18 g of water

Reason (R) in the combustion of methane, water is one of the product.

1) Both A and R are true but R is not the correct explanation of A

2) A is true but R is false

3) A is false but R is true

4) Both A and R are false

55. Given below are two statements

Statement-I: In octahedral complexes, when $\Delta_o < P$ high spin complexes are formed.

When $\Delta_o > P$ low spin complexes are formed.

Statement-II: In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed.in the light of above statements, choose the most appropriate answer from the options given below:

1) Statement I is correct but Statement II is incorrect

2) Statement I is incorrect but Statement II is correct

3) Both Statement I and Statement II are correct

4) Both Statement I and Statement II are incorrect

56. Match List-I with List -II

	List I		List-II
A	ICl	I	T-shape
B	ICl_3	II	Square pyramidal
C	ClF_5	III	Pentagonal bipyramidal
D	IF_7	IV	Linear

Choose the correct answer from the options given below.

1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

2)(A)-(I), (B)-(IV), (C)-(III), (D)-(II)

3) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

4) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

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57. $CaCO_3 \rightarrow CaO + CO_2$

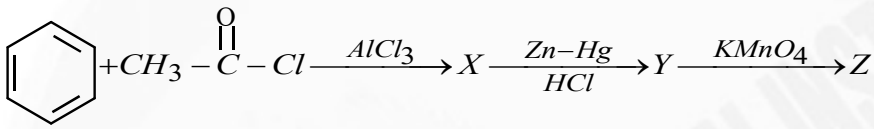
$CaO + C \rightarrow CaC_2 + CO$

$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$

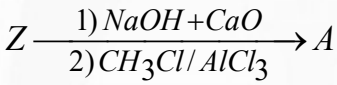
The weight of $CaCO_3$ needed for preparation of 0.224L of acetylene gas under STP(1atm and 273 K) is

1)10g **2)**1g **3)**5g **4)**2g

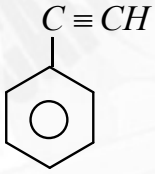
58.



$$+CH_3-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow{AlCl_3} X \xrightarrow[HCl]{Zn-Hg} Y \xrightarrow{KMnO_4} Z$$

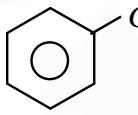


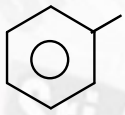
$$Z \xrightarrow[2)\,CH_3Cl/AlCl_3]{1)\,NaOH+CaO} A$$



A) X also formed by the, hydrolysis of $C \equiv CH$
[in the presence of $\%\,HgSO_4 + H_2SO_4$]

B) Y also formed by the, benzene reacts with ethyl chloride in presence of $AlCl_3$

C) Z also formed by the  with $KMnO_4 / H^{\oplus}$

D) 

$$+H_2N-NH_2 \xrightarrow{OH^-} A$$

Choose the most appropriate answer from the option given below

1) A, B and C only correct **2)** A, B only correct

3) A, B, C and D correct **4)** A, C only correct

59. Element of group-15 form compounds in +5 state, however bismuth forms only one well characterised compound in +5 oxidation state. The compound is

1) Bi_2O_5 **2)** BiF_5 **3)** $BiCl_5$ **4)** Bi_2S_5

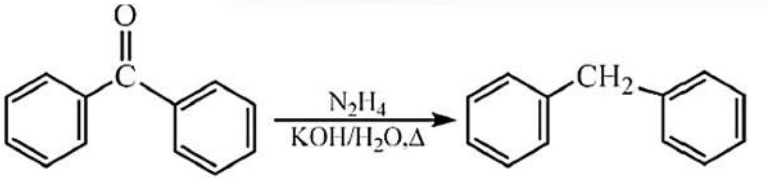
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60. The CFSE for octahedral $[CoCl_6]^{-4}$ is $18{,}000\,cm^{-1}$,the CFSE for tetrahedral $[CoCl_4]^{-2}$ will be (CFSE= crystal field stabilization energy)

1) $18000cm^{-1}$ **2)** $16000cm^{-1}$ **3)** $8000cm^{-1}$ **4)** $20000cm^{-1}$

61.



The reduction reaction is known as:

1) Roseenmund reduction **2)** Wolff-Kishner reduction

3) Stephen reduction **4)** Etard reduction

62. In a first order decomposition, the time taken for the decomposition of reactant to one fourth and one eight of its initial concentration are t_1 and t_2 respectively, the ratio t_1 / t_2 will be

1) $\frac{4}{3}$ **2)** $\frac{3}{2}$ **3)** $\frac{3}{4}$ **4)** $\frac{2}{3}$

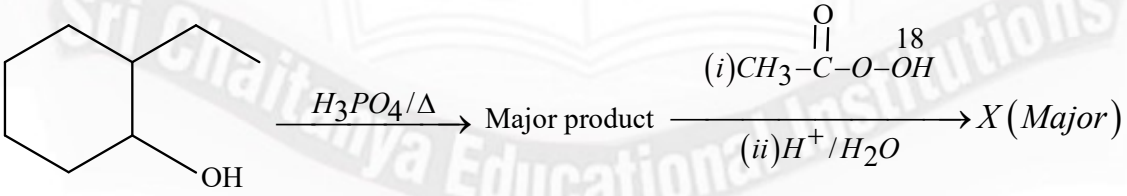
63. For the given reaction (1) and (2)

$p_{(g)} \rightleftharpoons Q_{(g)} + R_{(g)} \ldots\ldots(1)$

$A_{(g)} \rightleftharpoons 2B_{(g)} \ldots\ldots(2)$ (Given $K_{P_1} : K_{P_2} = 9:1$) if the degree of dissociation of $P_{(g)}$ *and* $A_{(g)}$ be the same then the total pressure at equilibrium (1) and (2) are in the ratio :

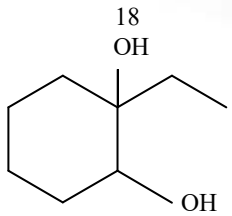
1) 3:1 **2)** 36:1 **3)** 1:1 **4)** 5:1

64.

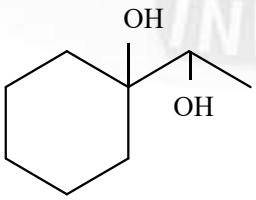


X is___

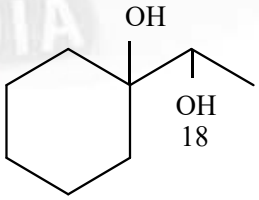
1)



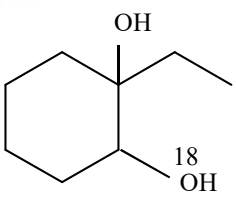
2)



3)



4)



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65. 1 mL of 0.01 N HCl is added to 999 mL solution of NaCl. The P^H of resulting solution will be

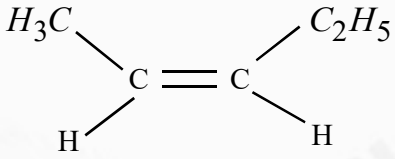
1) 8 **2)** 5 **3)** 3 **4)** 1

66. Which element -element bond has the highest dissociation energy?

1)C-C **2)**Si-Si **3)**Ge-Ge **4)**Sn-Sn

67. In the given reactions identify A and C.

$$A \xrightarrow[CH_3Br]{LiNH_2} B \xrightarrow{Pd/C}$$



$$B \xrightarrow{Na/Liquid\ NH_3} C$$

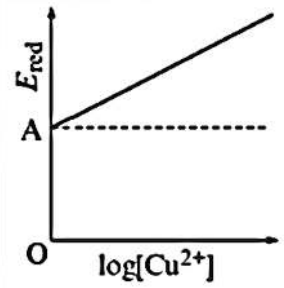
1) A:2-Butyne C: cis 2-pentene **2)** A:Butyne C:trans-2-pentene

3) A:2-pentyne C: cis -2-butene **4)** A:Butyne C: cis -2-pentene

68. The degree of dissociation of $Ca(NO_3)_2$ in a dilute aqueous solution, containing 7.0g of the salt per 100g water at $100^\circ C$ is 70%.if the Vapour pressure of water at $100^\circ C$ is 760mm Hg. Calculate vapour pressure of the solution.

1) 750.26 **2)**746.26 **3)**751.26 **4)**747.26

69. For the process: $Cu^{2+} + 2e^- \rightarrow Cu; \log\left[Cu^{2+}\right] vs. E_{red}$ graph is shown in the figure, where $OA = 0.34V$ .The electrode potential of the half-cell of $Cu\,|\,Cu^{2+}$ (0.1M) will be [2.303 RT/F=0.06]



1)-0.31 V **2)** +0.31 V **3)**-0.37V **4)** +0.37 V

70. What is the order of a reaction which has a rate expression rate=$k\left[A\right]^{3/2}\left[B\right]^{-1}$

1) 3/2 **2)** 1/2 **3)** 0 **4)** None of these

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Choose the number of CORRECT statements from the following

1) 1st ionisation potential of 'B' is higher that of 'Be'

2) Tl^{+} is more stable than Tl^{+3}

3) $[Ar]4s^{0}3d^{3}$ is the electronic configuration of V^{+3}

4) 2nd ionisation potential of $Na > 1^{st}$ ionisation potential of Na

5) 1st ionisation potential of N<2nd ionisation potential of N

6) Electronegativity of Cl> Electronegativity of F

7) Al is more atomic radius than Ga

8) Conversion of $O \rightarrow O^{-}$ is exothermic

9) The order of electronegativity of C, O, Si and P is $Si < P < C < O$

72. In the following reaction: $P_4 + xNaOH + yH_2O \rightarrow PH_3 + 3NaH_2PO_2$ Sum of x+y is_____

73. If ΔG^{0} for the half cell MnO_4^{-} / MnO_2 in an acid solution is $-xF$, then find the value of x

Given (Given: $E^{0}_{MnO_4^{-}/Mn^{2+}} = 1.5; E^{0}_{MnO_2/Mn^{2+}} = 1.25V$)

74. How many isomeric compound can have the formula $C_2ClFBrH$ _____

75. A sample of 0.50 g of an organic compound was treated according to Kjeldahls method. The ammonia evolved was absorbed in 50 ml of $0.5 MH_2SO_4$.the residual acid required 60 mL of 0.5 M solution of NaOH for neutralisation. Find the percentage composition of nitrogen in the compound.

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